

Chapter 8: STRUCTURAL BREAKS

- Occurrence of structural break means that “the world change”: changes can be legal, institutional, or geopolitical rules of the game.
- Our question is: How does structural breaks impact on unit root tests?
 - standard unit root tests do not take into account exogeneous shift and misrepresent the DGP
 - Perron (1989): if structural breaks are not taken into account, ADF tests are biased towards non-rejection. Using Nelson/Plosser(82) data, he concluded that macroeconomic variables are not generally characterized by unit root processes, but rather by structural breaks.
 - **Idea**: Random walk process: trend always changes. Rejection null of unit root: trend never changes. However, it would be better to compare “always” with “sometimes” and check the frequency of permanent shocks.
 - Important results in Perron(89,90): If there is a structural change (in slope; intercept) then standard unit root tests will have zero asymptotic power even if the null hypothesis of stationary is true.
 - Note: the KPSS test is also affected by the presence of break (reject too often).

STRUCTURAL BREAKS

- How to test for unit root when there is a structural change?
 - Vogelsang/Perron: systematic search of the break (endogenous approach): test for a unit root with unknown break dates (changes are allowed to occur under both the null and alternative hypothesis).
- Possible changes:
 1. Change in level for a non-trending series
 2. Change in level for a trending series;
 3. Change in slope
 4. Change in both, level and slope
- Different Transitions Effects (not only the transition is different, but the approach to test unit root is different):
 1. Additive Outlier Model (AO): change occurs quickly;
 2. Innovational Outlier model (IO): change to the new trend function is gradual

STRUCTURAL BREAKS

- A0 models (two-step approach):
 - Model (AO-0): change in level for a non-trend series (US interest rate):

$$y_t = \mu_1 + (\mu_1 - \mu_2)DU_t + \epsilon_t$$

- Model (AO-A): change in level for a trend series:

$$y_t = \mu_1 + \beta t + (\mu_1 - \mu_2)DU_t + \epsilon_t$$

- $DU_t = 1$ if $t > T_1$, and 0 otherwise. Where T_1 is the break date.
- $DT_t^* = t - T_1$ if $t > T_1$, and 0 otherwise.

- Model (AO-B): change is slope (useful for the analysis of post war quarterly real GNP for the G7 countries):

$$y_t = \mu_1 + \beta_1 t + (\beta_1 - \beta_2)DT_t^* + \epsilon_t$$

- Model (AO-C): change in both, level and slope:

$$y_t = \mu_1 + \beta t + (\mu_1 - \mu_2)DU_t + (\beta_1 - \beta_2)DT_t^* + \epsilon_t$$

- $DU_t = 1$ if $t > T_1$, and 0 otherwise. Where T_1 is the break date.
- $DT_t^* = t - T_1$ if $t > T_1$, and 0 otherwise.

Note: changes in the trend function are allowed to occur under both, the null and alternative hypotheses.

- $\Delta\epsilon_t = C(L)e_t$, where e_t is a white noise.

Testing if there is a unit root when the trend is not stationary

- Procedure:
 - Detrend the data (estimate under the alternative hypothesis): use the residuals ($\tilde{y}_t$)
 - Then test the residual of the estimated equation (note about including)
 - For the AO models AO-0, AO-A and AO-C we need to include the present and past of the changing in level dummies (For AO-B we do not need to include it due no change in level; the two segments of the trend are joined at the time of the break: the limit distribution is different, see Perron/Vogelsang(1993).

$$\tilde{y}_t = \alpha \tilde{y}_{t-1} + \sum_{j=0}^k d_j D(T_1)_{t-j} + \sum_{i=1}^k \gamma_i \Delta \tilde{y}_{t-i} + \epsilon_t$$

- Where:
 - $D(T_1)_t = 1$ if $t = T_1 + 1$, and 0 otherwise.

Perron (89) showed that the current value and lags of the dummy for changing in level should be included in the ADF test, so the limit distribution of the tests are the same as for the IO case.

The critical values depend on the occurrence of the break relative to the sample size (critical values can be found on Perron(89)).

- IO models (nested models): The IO models are described separated, under the null and under the alternative hypotheses (IO-UR and IO-TS):
 - Model (IO-0-UR): shows the immediate and the long-run impact of the change in the intercept – non-trend series:

$$y_t = y_{t-1} + C(L)(e_t + \delta D(T_1)_t)$$

- Model (IO-A-UR): shows the immediate and the long-run impact of the change in the intercept – trend series:

$$y_t = y_{t-1} + b + C(L)(e_t + \delta D(T_1)_t)$$

- Model (IO-C-UR): shows the immediate and the long-run impact of the change in intercept and in the slope:

$$y_t = y_{t-1} + b + C(L)(e_t + \delta D(T_1)_t + \eta DU_t)$$

- Where:
 - $DU_t = 1$ if $t > T_1$, and 0 otherwise. Where T_1 is the break date.
 - $D(T_1)_t = 1$ if $t = T_1 + 1$, and 0 otherwise.

- Model (IO-0-TS): shows the immediate and the long-run impact of the change in the intercept – non-trend series:

$$y_t = \mu + C(L)^*(e_t + \theta DU_t)$$

- Model (IO-A-TS): shows the immediate and the long-run impact of the change in the intercept – trend series:

$$y_t = \mu + \beta t + C(L)^*(e_t + \theta DU_t)$$

- Model (IO-C-TS): shows the immediate and the long-run impact of the change in intercept and in the slope:

$$y_t = \mu + \beta t + C(L)^*(e_t + \theta DU_t + \gamma DT_t^*)$$

where: $C(L)^* = (1 - L)^{-1}C(L)$

- The IO can be nested model to be tested under the null and alternative hypotheses:

$$y_t = \mu + \theta DU_t + \beta t + \gamma DT_t^* + \alpha y_{t-1} + \sum_{i=1}^k c_i \Delta y_{t-i} + \delta D(T_1)_t + \epsilon_t$$

- Where k is chosen by using Ng-Perron sequential test procedure.
- Under the null:

- IO-A: $\alpha=1; \beta = \theta = 0; \delta \neq 0$
- IO-: $\alpha=1; \beta = \gamma = 0; \delta \neq 0$

- Under the alternative:

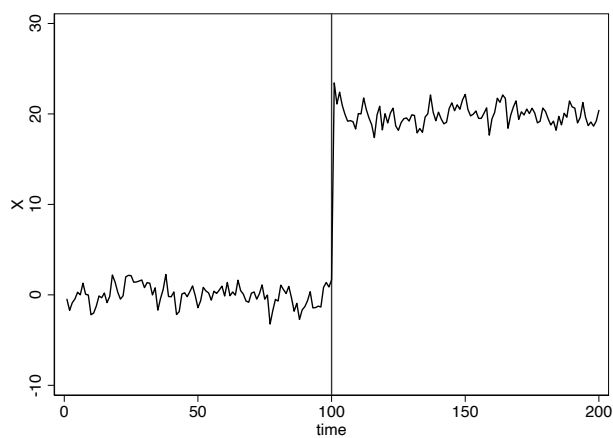
- $|\alpha| < 1$; in general $\delta = 0$

- Perron/Vogelsang (92): the choice of break is based on the minimum value of the t-stat that $\alpha = 1$ (like Zivot-Andrews) and also based on the maximum (absolute) values of the t-stat on the coefficient of the appropriate shift dummy (more powerful test).
- Vogelsang/Perron (98) showed that if the noise component has a unit root and breaks occurs in the trend function, Zivot/Andrews (92) tests often diverge or are not invariant to break parameters (substantially less power than Perron (89) tests).
- Critical values of the unit root test will depend on the determinist trend specified. Kim-Perron (2009) developed new test procedure, allowing for a break in the trend function at unknown time under both the null and alternative hypotheses. Their approach improve the power of the unit root test and maintain the correct size. Their test is based on a pre-test for a change in the trend function: Perron-Yabu (2009) test.
- Campbell-Perron: it is important to correctly identify the model. Otherwise the power of the test goes to zero. Guidelines: always plot your data.

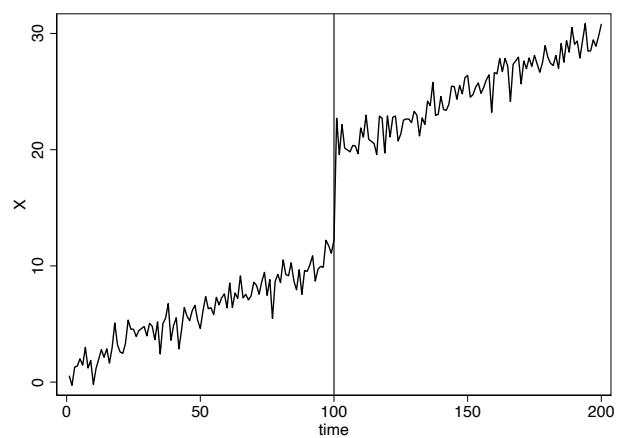
a) Change in intercept (not trend)

b) Change in intercept (trend stationary)

(a)

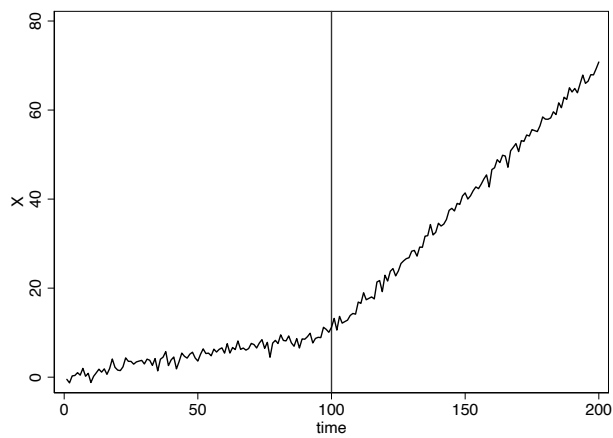


(b)

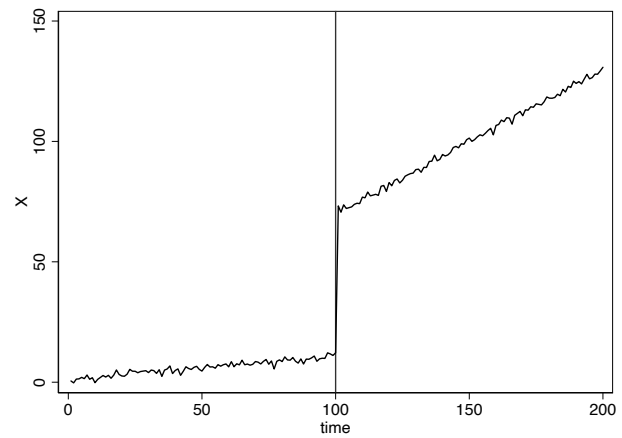


- c) Change in slope (spliced together)
- d) Change in intercept and slope

(c)



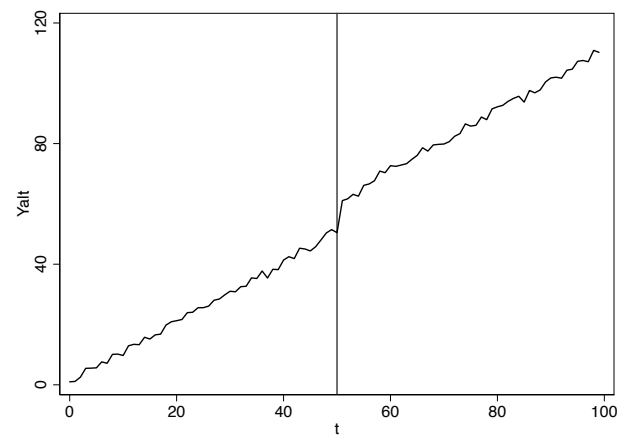
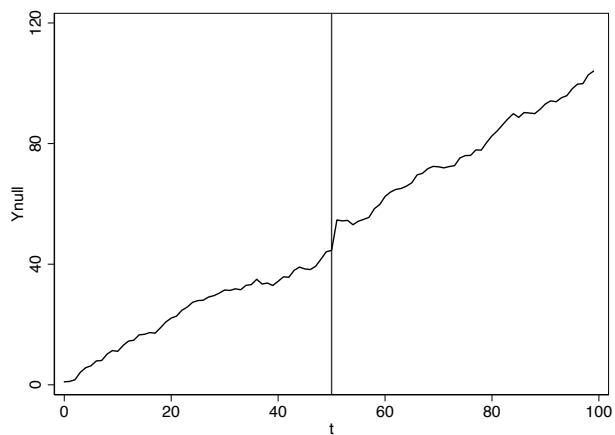
(d)



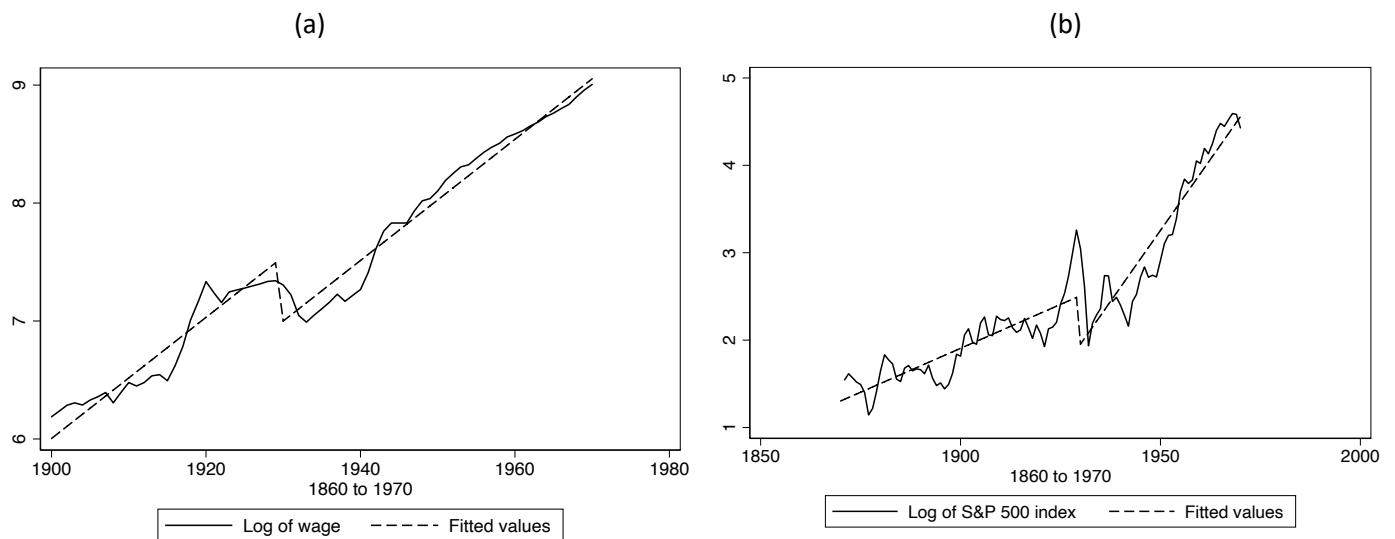
$$H_0: y_t = \beta_0 + y_{t-1} + \beta_2 DU_t + \epsilon_t$$

$$H_A: y_t = \beta_0 + \beta_1 y_{t-1} + \beta_2 DU_t + \epsilon_t$$

Testing if there is a change in the intercept but no change in slope



Perron (89): Tests for a Unit Root with a Known Structural Break



```
cd "/Users/catir/Desktop/"
use "NelsonPlosserData.dta", clear
```

```
* Figure a
```

```
    gen DU = 0
    replace DU = 1 if year > 1929
    quietly reg lwg DU year
    predict yhat
    graph twoway (connected lwg year if (year>=1900 & year <=1970), msymbol(i)) (connected yhat year if (year>=1900 &
year <=1970), msymbol(i))
        graph save "/Users/perron/Desktop/book-do-files-for-distribution (3)/figure8.5a.gph" , replace
    drop yhat
```

```
* Figure b
```

```
    gen DT = 0
    replace DT = year if year > 1929
    quietly reg lsp500 DU year DT
    predict yhat
    graph twoway (connected lsp500 year if (year>=1870 & year <=1970), msymbol(i)) (connected yhat year if (year>=1870 &
year <=1970), msymbol(i))
        graph save "/Users/catir/Desktop/book-do-files-for-distribution (3)/figure8.5b.gph" , replace
    drop yhat
```

Perron's Results

- The code (posted on blackboard learn) will allow you to reproduce Perron's results (The Great Crash, the Oil Shock, and the Unit Root Hypothesis. Econometrica) for Nelson-Plosser data.
- To test if α is statistically significant from 1, we need different critical values than the ones given by STATA.
- Critical values depend on the position of the breaks relative to the sample, among other things (if no break, then the critical values are the ones given by STATA) . The critical values are found in Perron(1989).
- Perron's results are quite different than Nelson-Plosser (1982).
- Not allowing for structural breaks induces bias in the standard unit root test, making difficult to reject the null hypothesis.
- However, Perron(1989) assumed that the date of the break was known. But due to the uncertainty, breaks need to be estimated when they are unknown.

Zivot and Andrews's Test of a break at an unknown data

- Zivot and Andrews (92) consider the following IO type specification:

$$y_t = \mu + \theta DU_t + \beta t + \gamma DT_t^* + \alpha y_{t-1} + \sum_{i=1}^k c_i \Delta y_{t-i} + \epsilon_t$$

- $DU_t = 1$ if $t > T_1$, and 0 otherwise. Where T_1 is the break date.
- $DT_t^* = t - T_1$ if $t > T_1$, and 0 otherwise.

The break is chosen endogenously by minimizing the value of the t-statistic for testing $\alpha = 1$ over all possible break dates (data are trimmed).

Zivot and Andrew's Test of a break at an unknown data

- Zivot and Andrews (1992): test for a break point at an unknown date. The test is done for each possible break (each year); the break date is then chosen as the one that gives the most weight to the alternative hypothesis (the trend-stationary). They take the minimum value of the ADF-type unit root test.
- Zivot and Andrews' procedure can only identify the existence of a single structural breaks. They use the sequential Ng-Perron test to choose k-lags.
- Zivot and Andrews results rely on the assumption that structural breaks and unit roots are strict substitutes (no break under the null hypothesis).
- **Problem with Zivot and Andrews approach:** Vogelsang and Perron (98) showed that the limit result of Zivot-Andrews test is not valid if a break is present under the null. They showed that the test often diverge to minus infinite when a shift in slope is present (this implies that a rejection can be due to the presence of a unit root process with a breaking trend. It has substantially less power than Perron(89)

Zivot and Andrews' command on STATA

- If you have not installed yet the command, just search on STATA for:
`zandrews` command and click install.

To run, type:

```
zandrews varname, maxlags(#) break(here is the type of break: it can be intercept, trend, or  
both) level(here is the level of significance used to choose the break, default is 0.10)  
lagmethod(AIC, BIC or TTest options)
```

The procedure can choose the lags by using the sequential t-test (Ng-Perron) or AIC, BIC criteria. However, it is well known that the sequential t-test improves the power of the unit root test. We should use the sequential t-test to choose the the lags.

Kim-Perron (2009) test

- Perron (89) tests are invariant to parameters of the trend function (when breaks are known). Kim-Perron(2009) developed a new procedure, which allows for a break in the trend function at an unknown time under both the null and alternative hypotheses. This allows for an increase in power as the same time maintain a correct size, even in finite samples. Their procedure is not an extension of Zivot-Andrews test. Their test is based on a pre-test for a change in the trend function, known as Yabu-Perron(2009) test. If the pre-test rejects, then the limit distribution of the unit-root test is the same as if a break date was known. Their limit distribution can be found in Perron(89).
- The code for both, Yabu-Perron and Kim-Perron tests can be found on Perron's web page (search for codes); both can be run on Matlab.